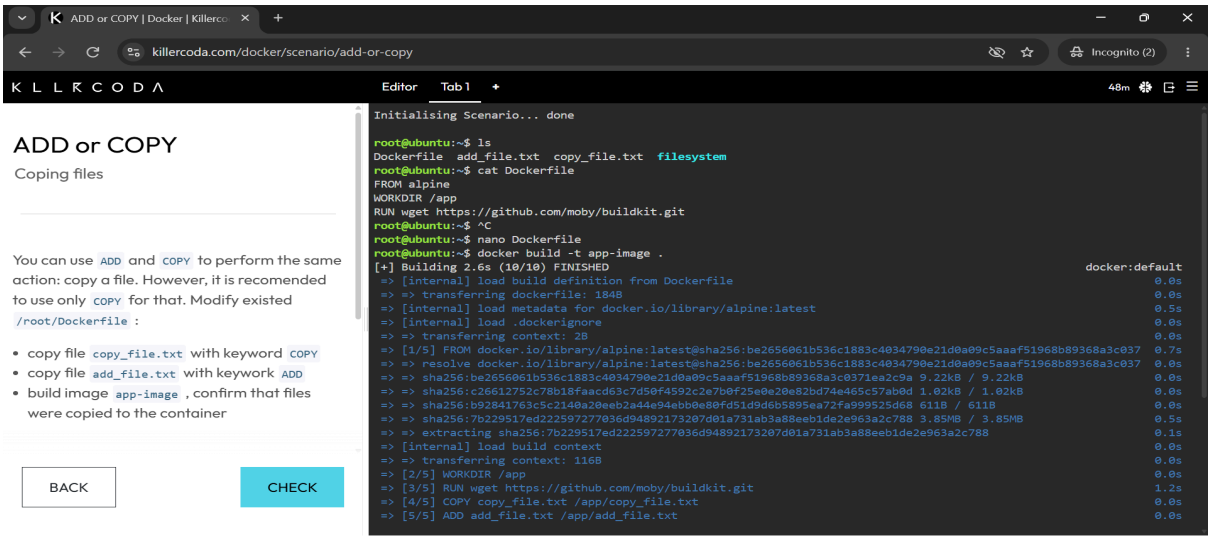
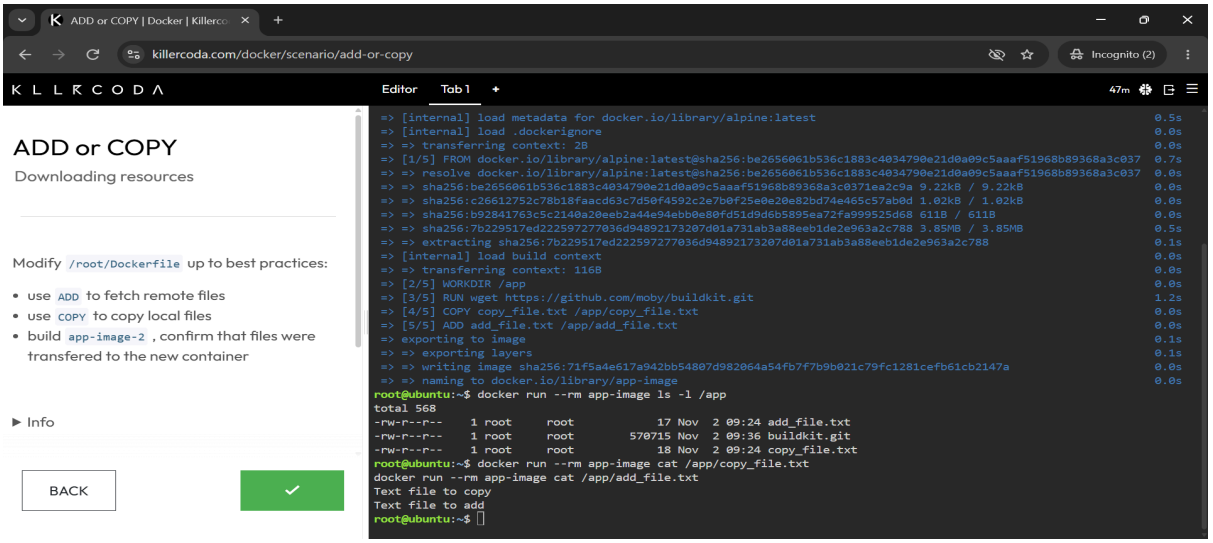
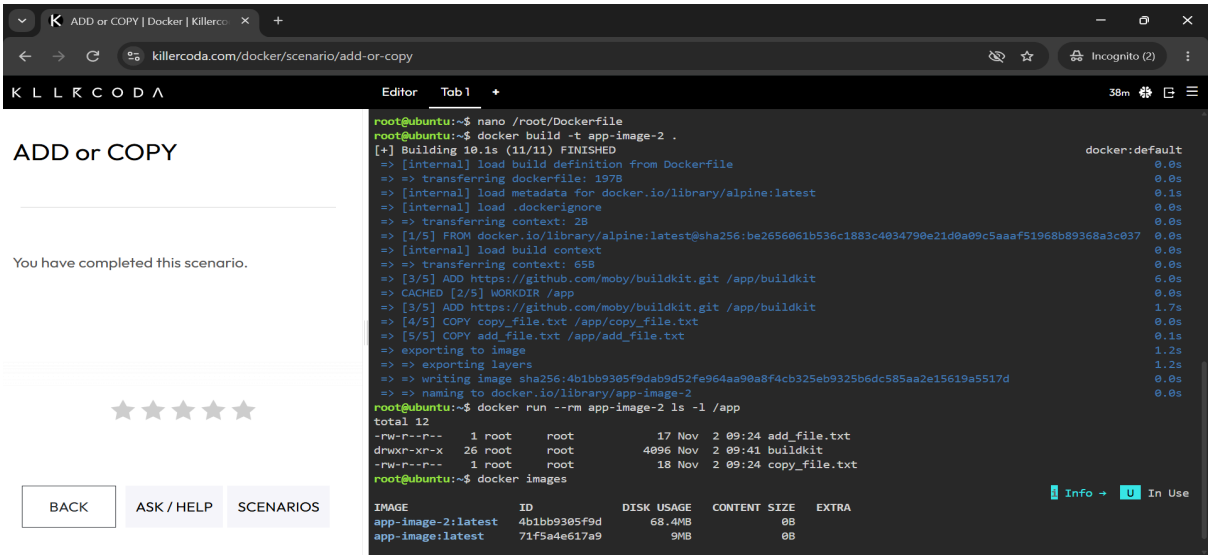
Docker Assignment

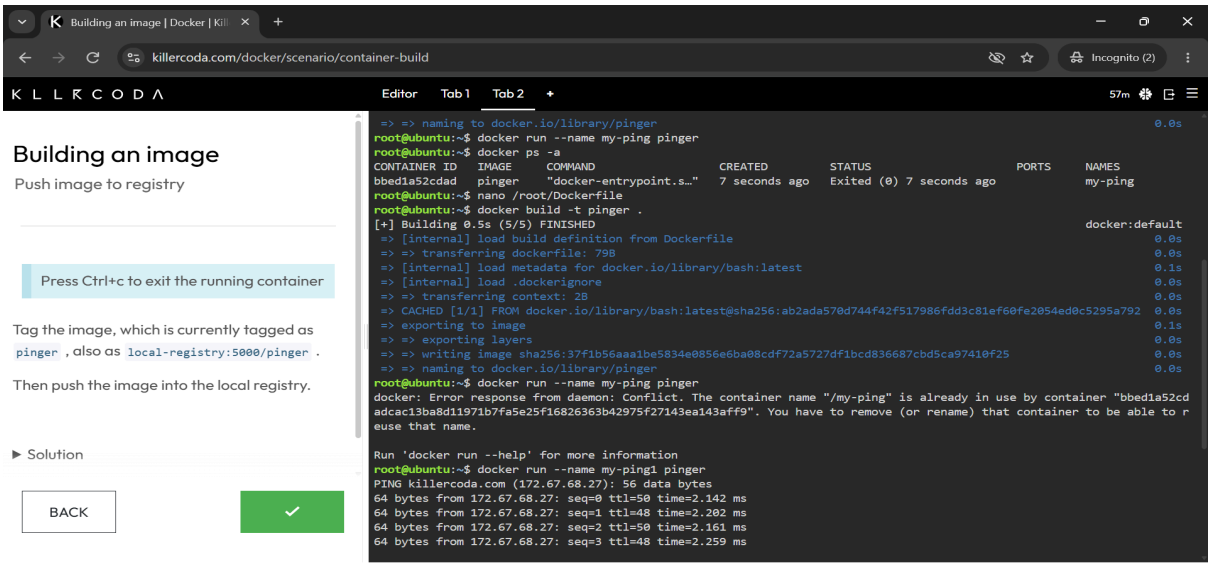
1. Basics

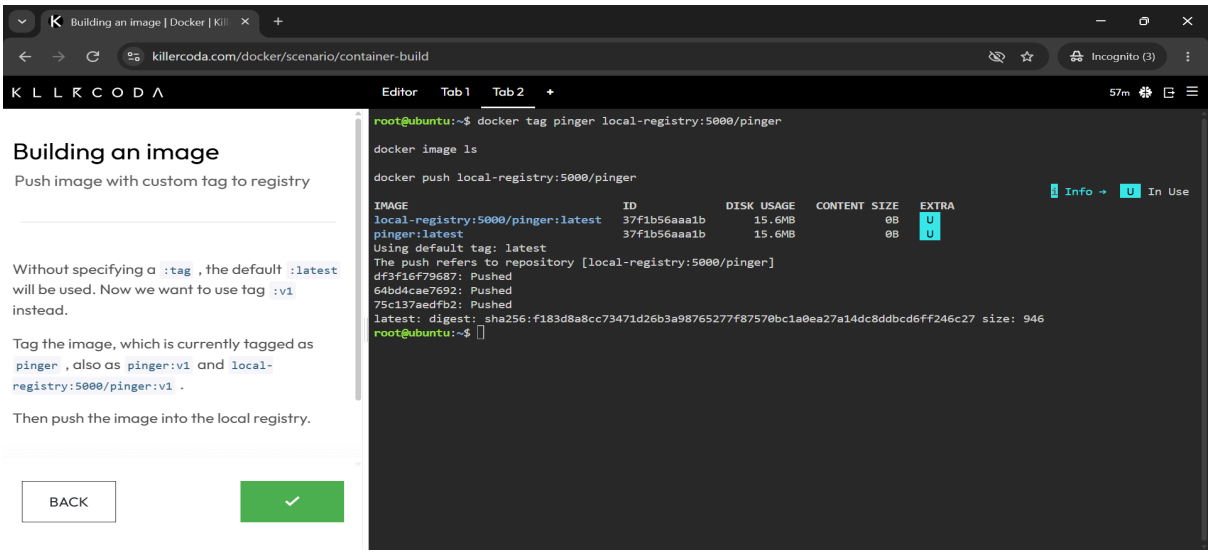


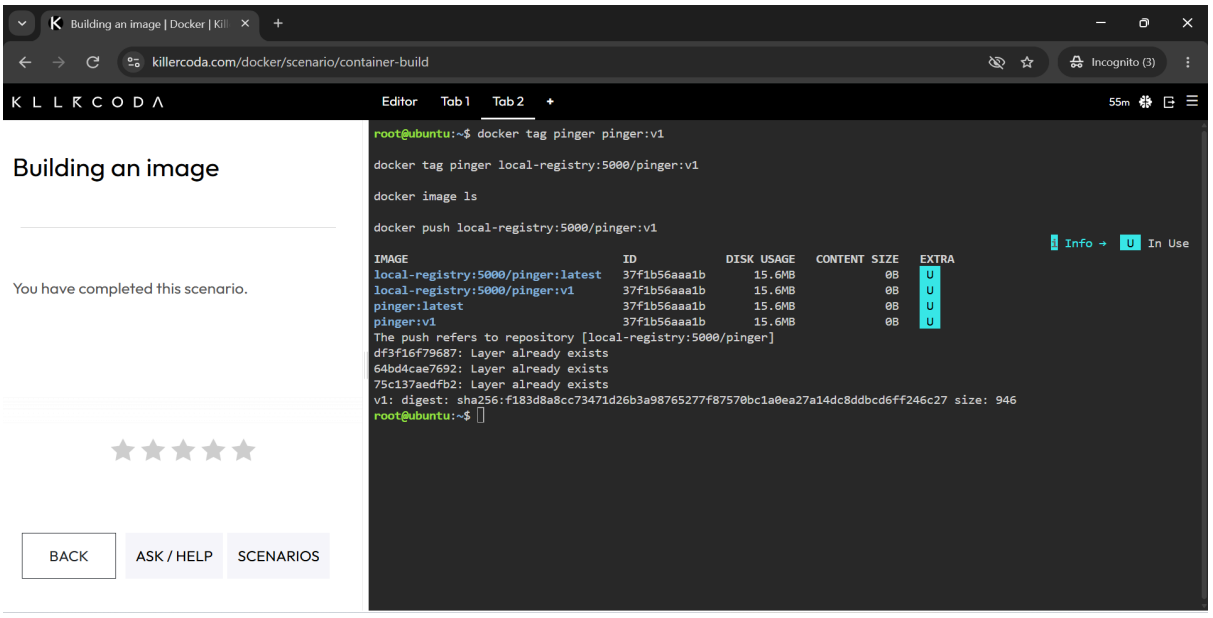




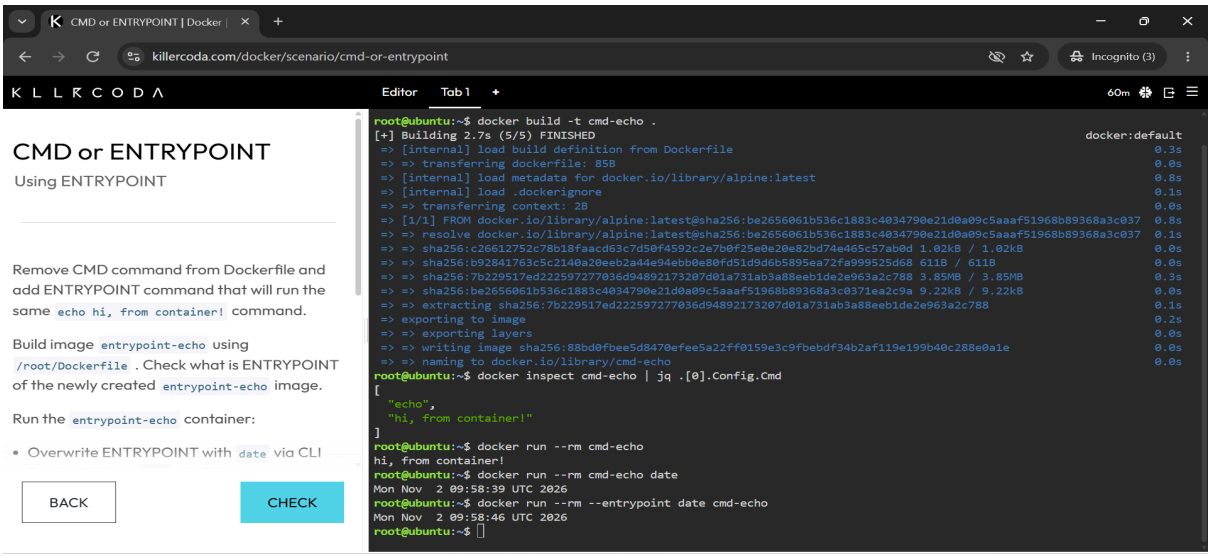
2. Image Building

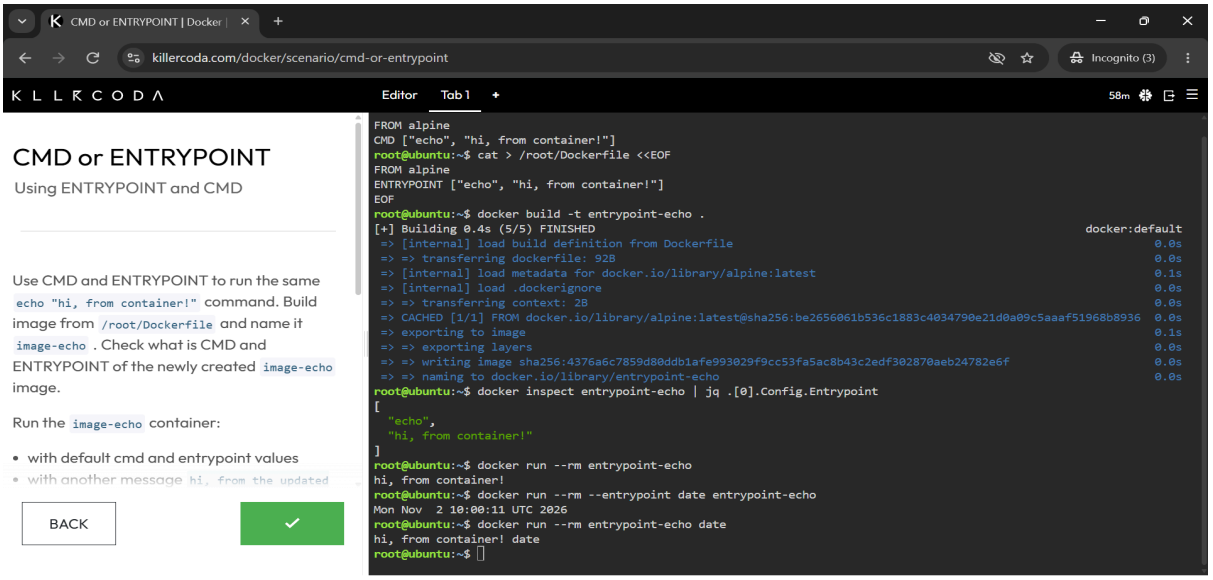


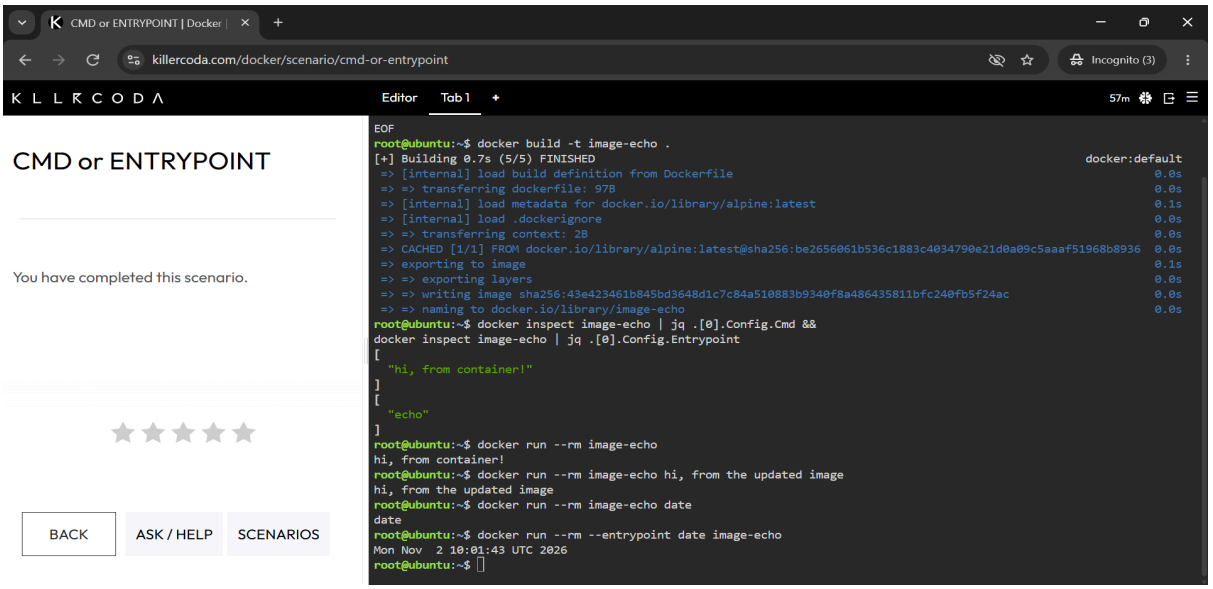




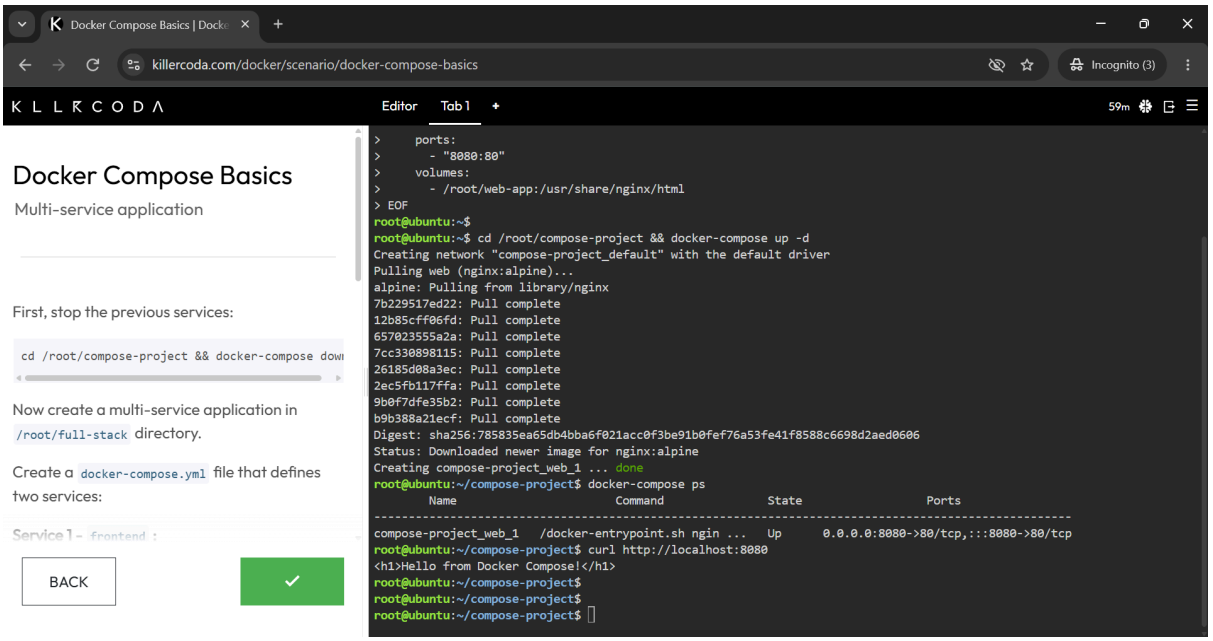
3. CMD/Entrypoint

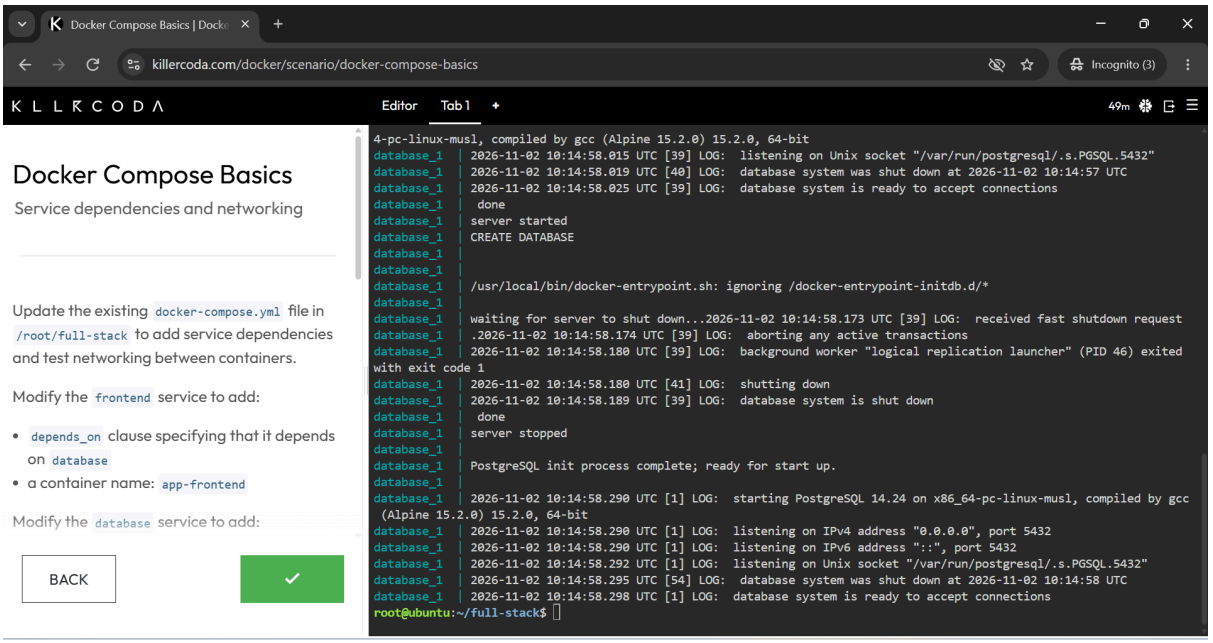


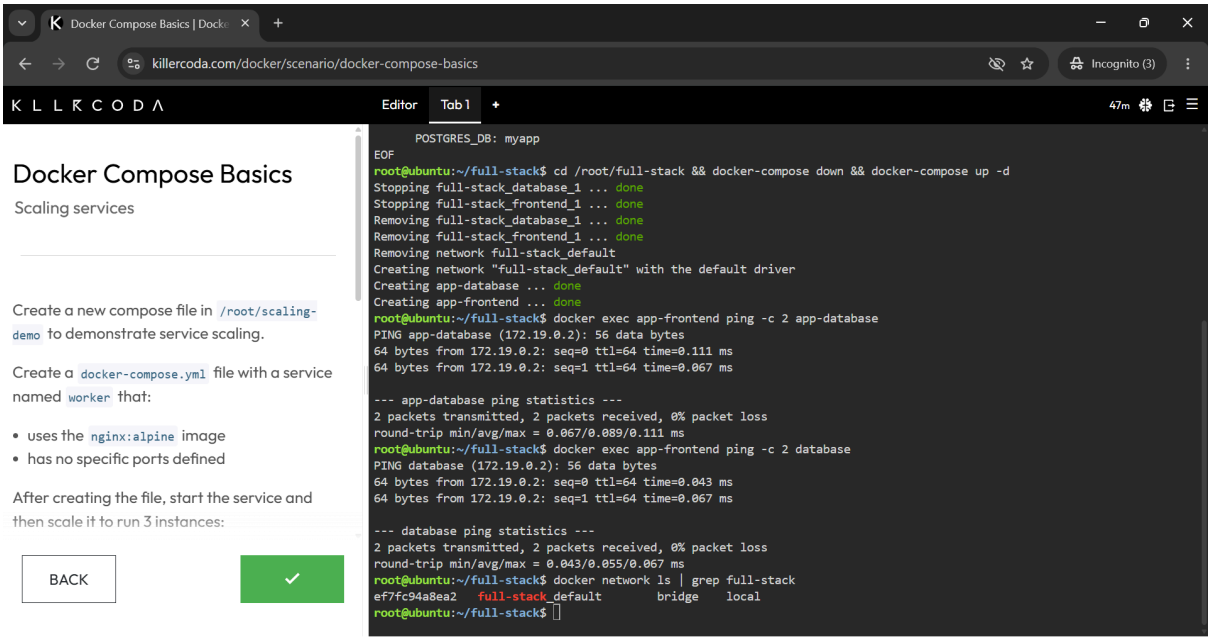


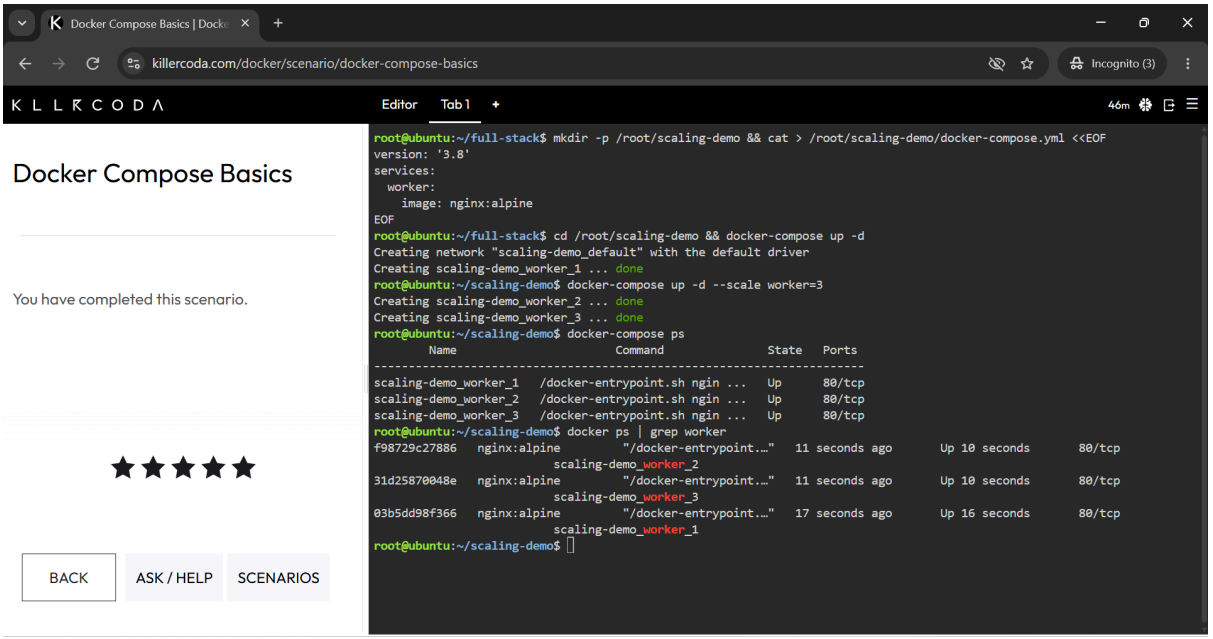


4. Docker Compose

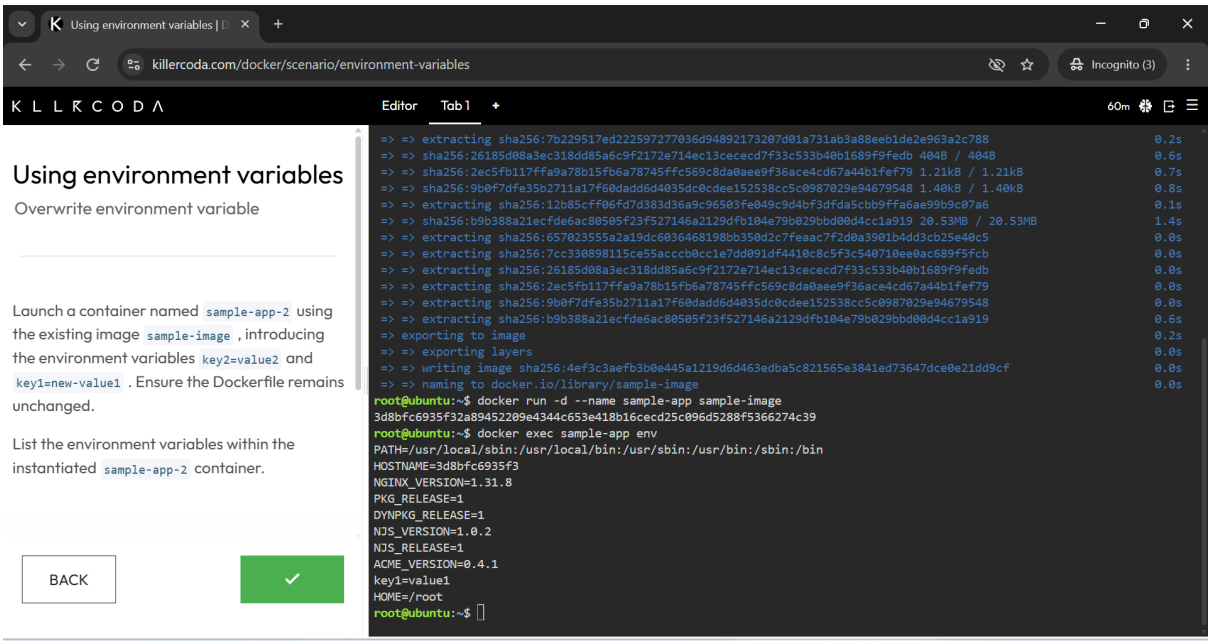


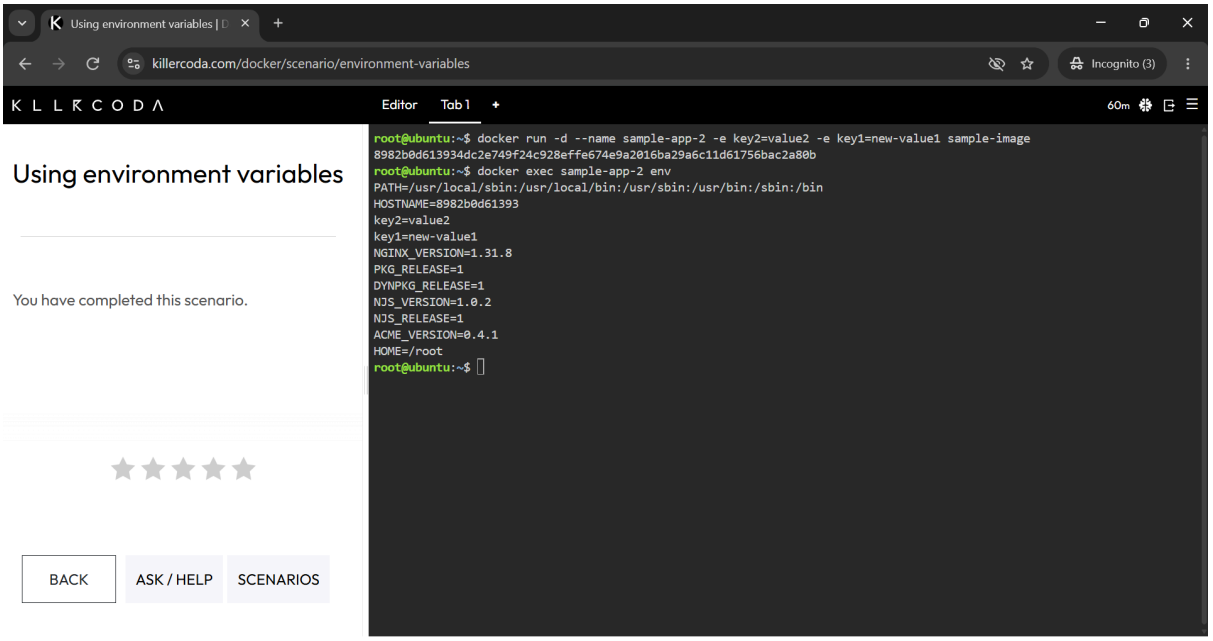



Docker Compose Basics | Docke
killercoda.com/docker/scenario/docker-compose-basics
Incognito (3)
KILLERCODA
Editor
Tab 1
47m
Docker Compose Basics
Scaling services
Create a new compose file in /root/scaling-demo to demonstrate service scaling.
Create a docker-compose.yml file with a service named worker that:
uses the nginx:alpine image
has no specific ports defined
After creating the file, start the service and then scale it to run 3 instances:
BACK
POSTGRES_DB: myapp
EOF
root@ubuntu:~/full-stack$ cd /root/full-stack && docker-compose down && docker-compose up -d
Stopping full-stack_database_1 ... done
Stopping full-stack_frontend_1 ... done
Removing full-stack_database_1 ... done
Removing full-stack_frontend_1 ... done
Removing network full-stack_default
Creating network "full-stack_default" with the default driver
Creating app-database ... done
Creating app-frontend ... done
root@ubuntu:~/full-stack$ docker exec app-frontend ping -c 2 app-database
PING app-database (172.19.0.2): 56 data bytes
64 bytes from 172.19.0.2: seq=0 ttl=64 time=0.111 ms
64 bytes from 172.19.0.2: seq=1 ttl=64 time=0.067 ms
--- app-database ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 0.067/0.089/0.111 ms
root@ubuntu:~/full-stack$ docker exec app-frontend ping -c 2 database
PING database (172.19.0.2): 56 data bytes
64 bytes from 172.19.0.2: seq=0 ttl=64 time=0.043 ms
64 bytes from 172.19.0.2: seq=1 ttl=64 time=0.067 ms
--- database ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 0.043/0.055/0.067 ms
root@ubuntu:~/full-stack$ docker network ls | grep full-stack
ef7fc94a8ea2 full-stack_default bridge local
root@ubuntu:~/full-stack$

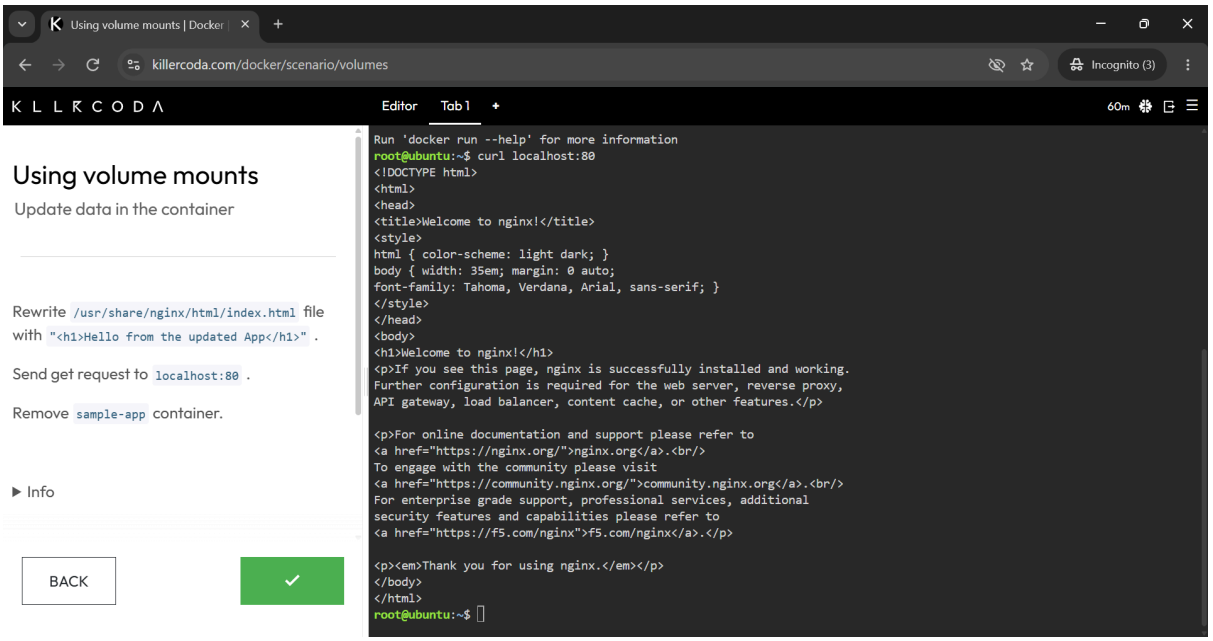

Docker Compose Basics | Docke
killercoda.com/docker/scenario/docker-compose-basics
Incognito (3)
KILLERCODA
Editor
Tab 1
46m
Docker Compose Basics
You have completed this scenario.
BACK
ASK / HELP
SCENARIOS
root@ubuntu:~/full-stack$ mkdir -p /root/scaling-demo && cat > /root/scaling-demo/docker-compose.yml <<EOF
version: '3.8'
services:
worker:
image: nginx:alpine
EOF
root@ubuntu:~/full-stack$ cd /root/scaling-demo && docker-compose up -d
Creating network "scaling-demo_default" with the default driver
Creating scaling-demo_worker_1 ... done
root@ubuntu:~/scaling-demo$ docker-compose up -d --scale worker=3
Creating scaling-demo_worker_2 ... done
Creating scaling-demo_worker_3 ... done
root@ubuntu:~/scaling-demo$ docker-compose ps
Name Command State Ports
scaling-demo_worker_1 /docker-entrypoint.sh ngin ... Up 80/tcp
scaling-demo_worker_2 /docker-entrypoint.sh ngin ... Up 80/tcp
scaling-demo_worker_3 /docker-entrypoint.sh ngin ... Up 80/tcp
root@ubuntu:~/scaling-demo$ docker ps | grep worker
f98729c27886 nginx:alpine "/docker-entrypoint.…" 11 seconds ago Up 10 seconds 80/tcp scaling-demo_worker_2
31d25870048e nginx:alpine "/docker-entrypoint.…" 11 seconds ago Up 10 seconds 80/tcp scaling-demo_worker_3
03b5dd98f366 nginx:alpine "/docker-entrypoint.…" 17 seconds ago Up 16 seconds 80/tcp scaling-demo_worker_1
root@ubuntu:~/scaling-demo$

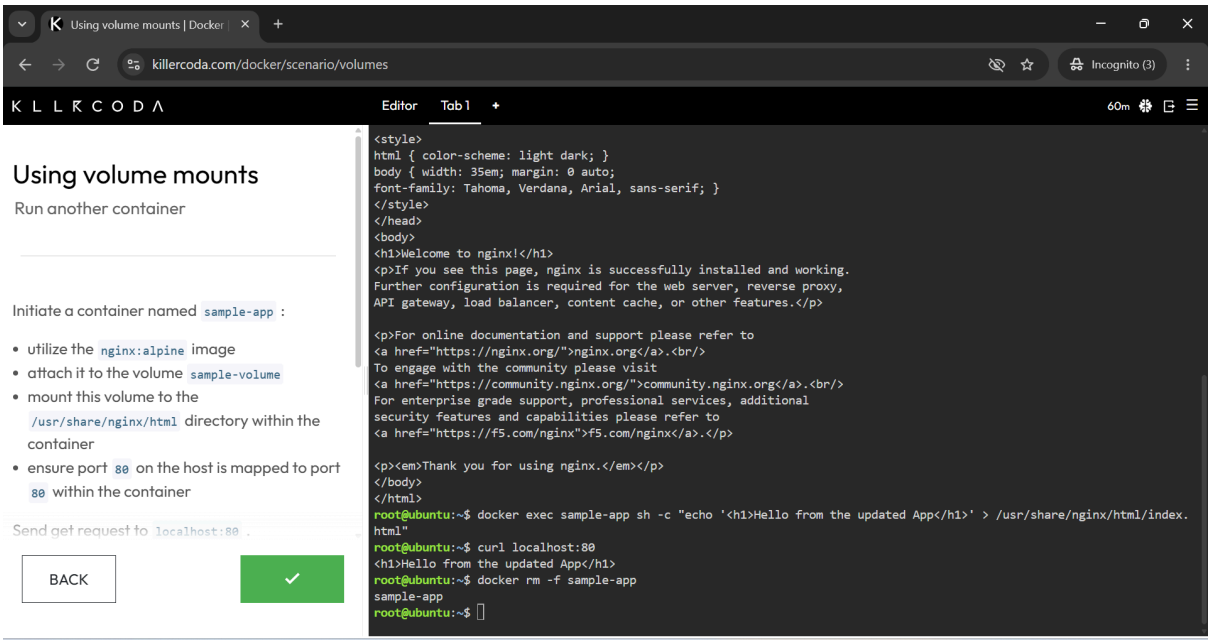
5. Environment Variables

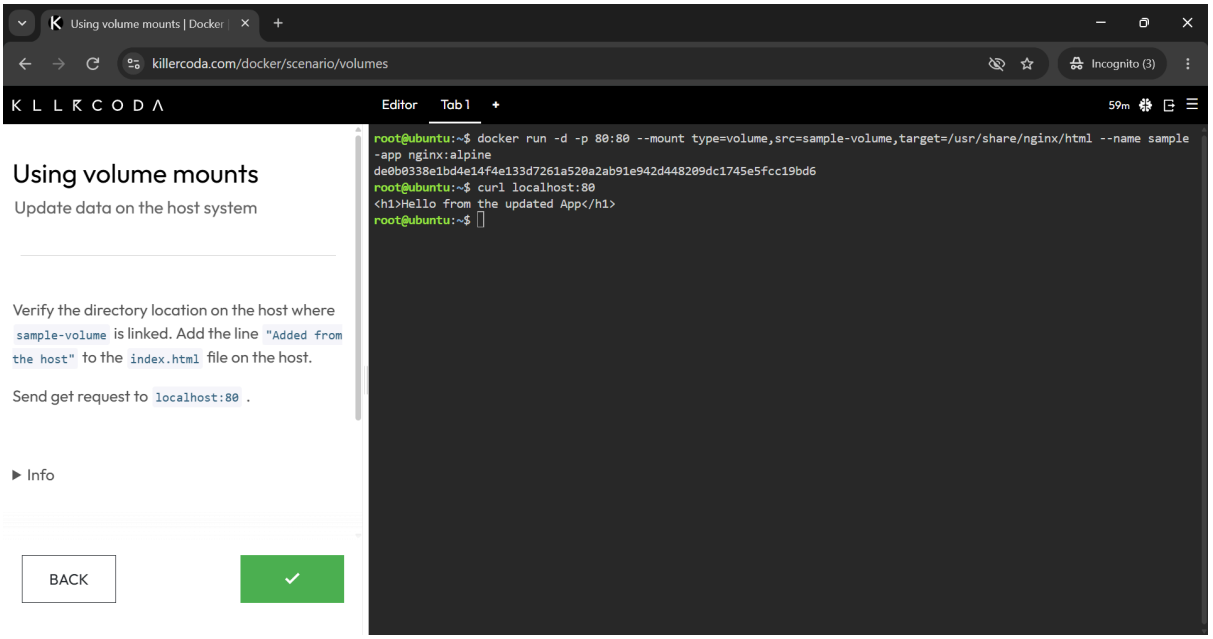


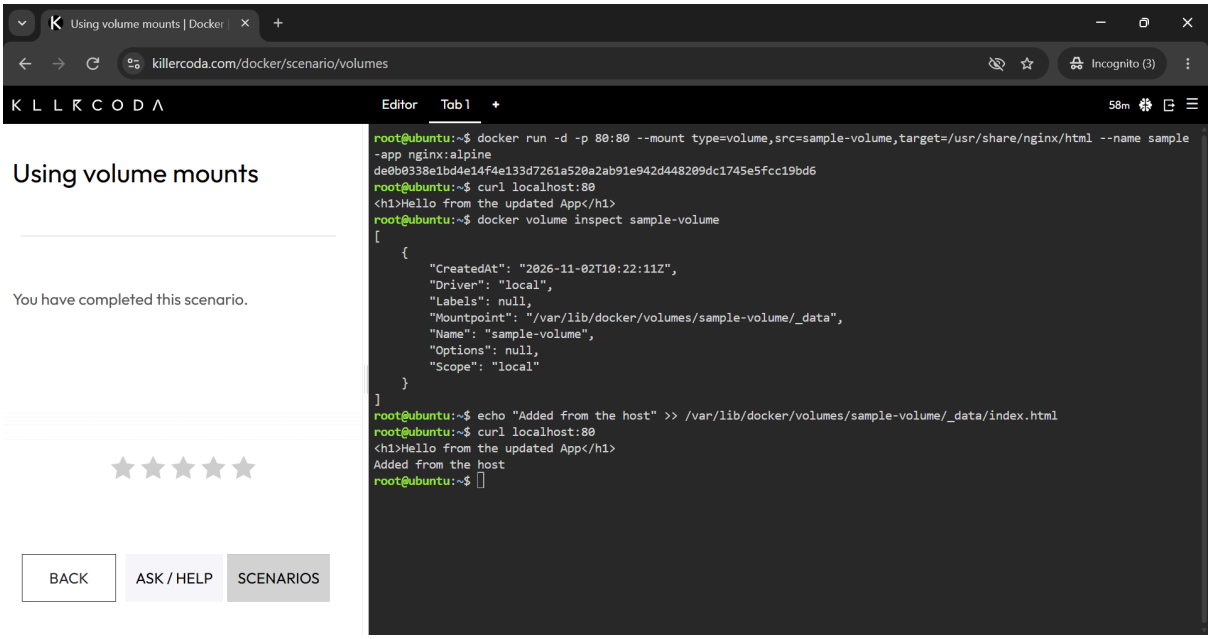


6. Volume mounts

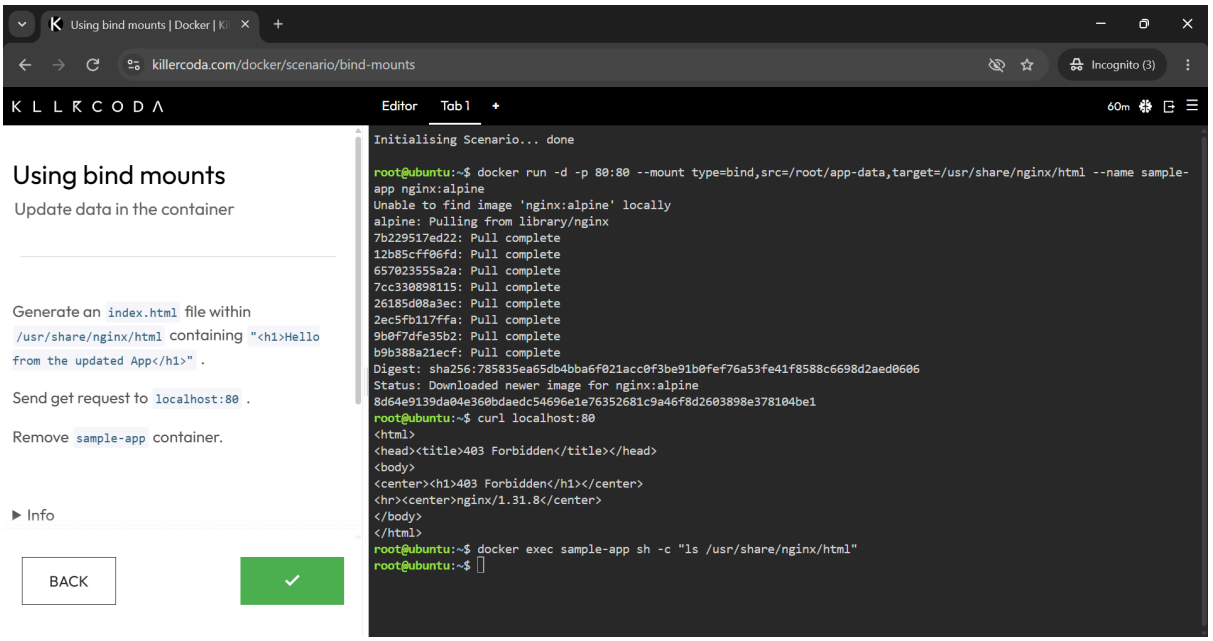


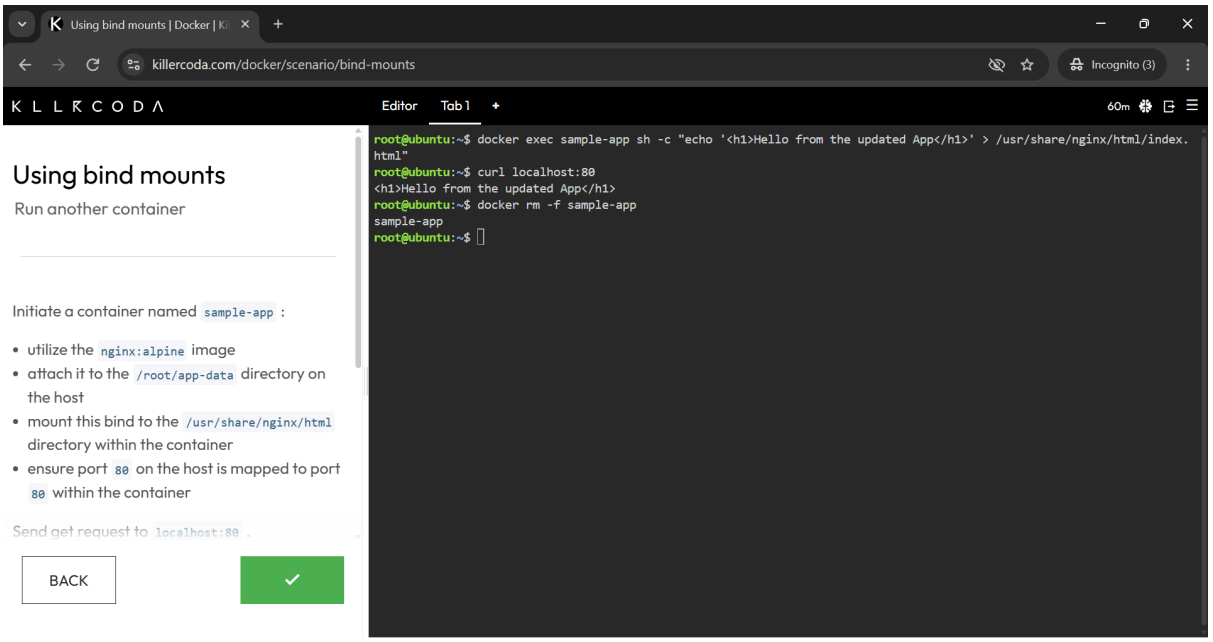


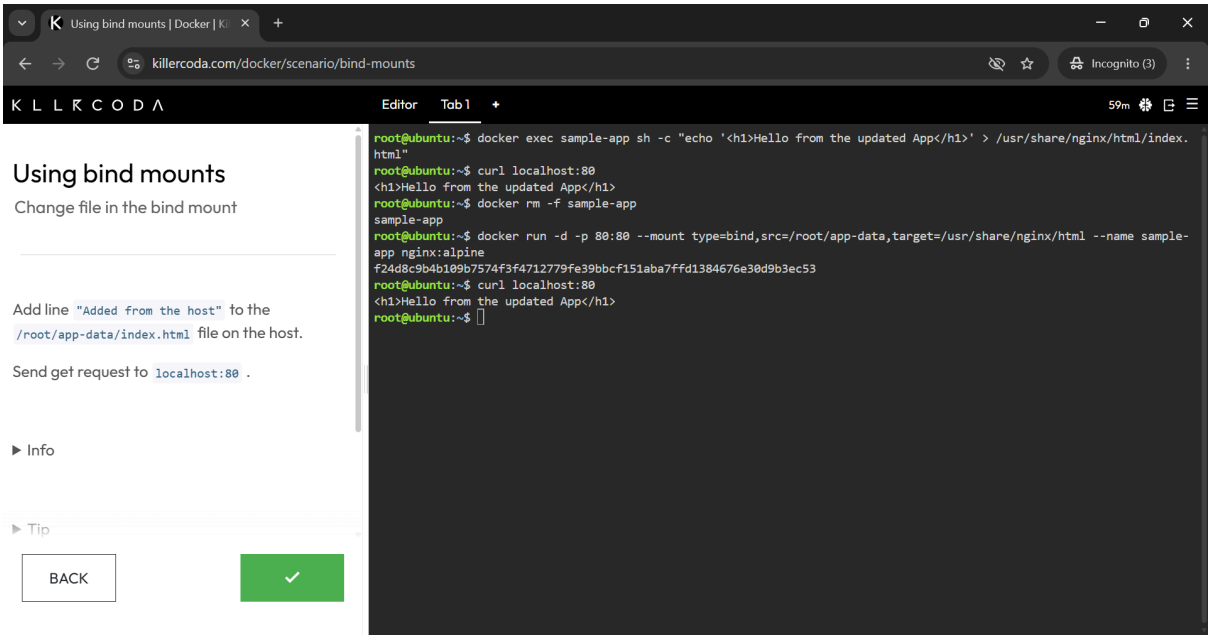

Using volume mounts | Docker
killercoda.com/docker/scenario/volumes
Incognito (3)
KILLERCODA
Editor
Tab 1
59m
Using volume mounts
Update data on the host system
Verify the directory location on the host where sample-volume is linked. Add the line "Added from the host" to the index.html file on the host.
Send get request to localhost:80 .
Info
BACK
root@ubuntu:~$ docker run -d -p 80:80 --mount type=volume,src=sample-volume,target=/usr/share/nginx/html --name sample-app nginx:alpine
de0b0338e1bd4e14f4e133d7261a520a2ab91e942d448209dc1745e5fcc19bd6
root@ubuntu:~$ curl localhost:80
<h1>Hello from the updated App</h1>
root@ubuntu:~$

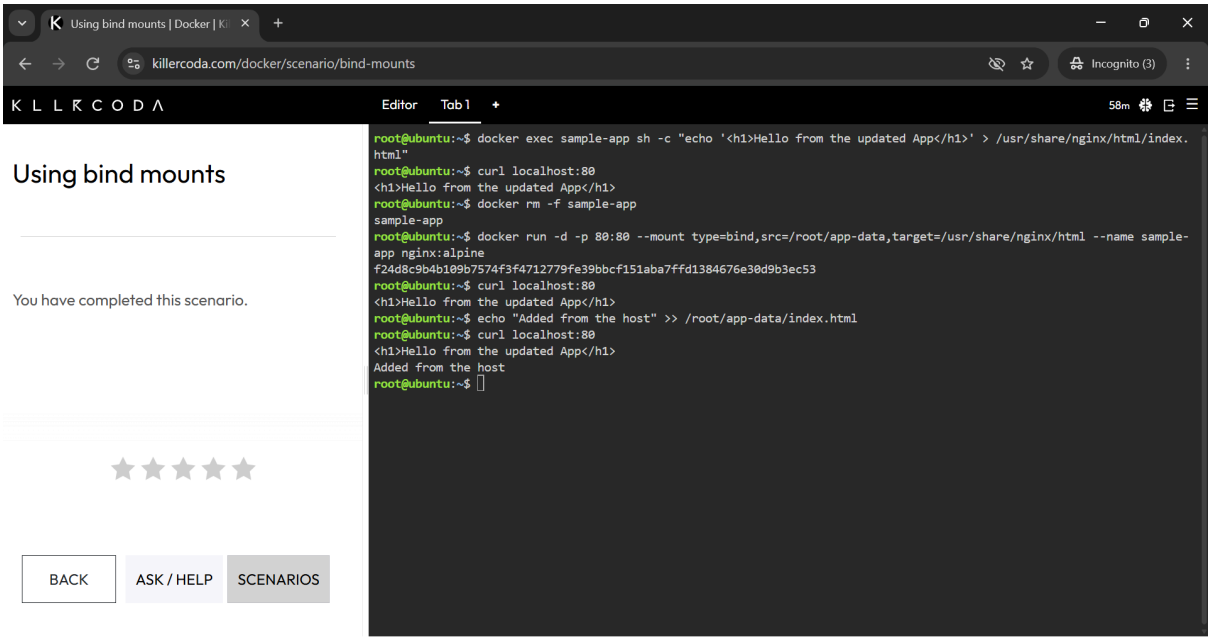

Using volume mounts | Docker
killercoda.com/docker/scenario/volumes
Incognito (3)
KILLERCODA
Editor
Tab 1
58m
Using volume mounts
You have completed this scenario.
BACK
ASK / HELP
SCENARIOS
root@ubuntu:~$ docker run -d -p 80:80 --mount type=volume,src=sample-volume,target=/usr/share/nginx/html --name sample-app nginx:alpine
de0b0338e1bd4e14f4e133d7261a520a2ab91e942d448209dc1745e5fcc19bd6
root@ubuntu:~$ curl localhost:80
<h1>Hello from the updated App</h1>
root@ubuntu:~$ docker volume inspect sample-volume
[
{
"CreatedAt": "2026-11-02T10:22:11Z",
"Driver": "local",
"Labels": null,
"Mountpoint": "/var/lib/docker/volumes/sample-volume/_data",
"Name": "sample-volume",
"Options": null,
"Scope": "local"
}
]
root@ubuntu:~$ echo "Added from the host" >> /var/lib/docker/volumes/sample-volume/_data/index.html
root@ubuntu:~$ curl localhost:80
<h1>Hello from the updated App</h1>
Added from the host
root@ubuntu:~$

7. Bind mounts

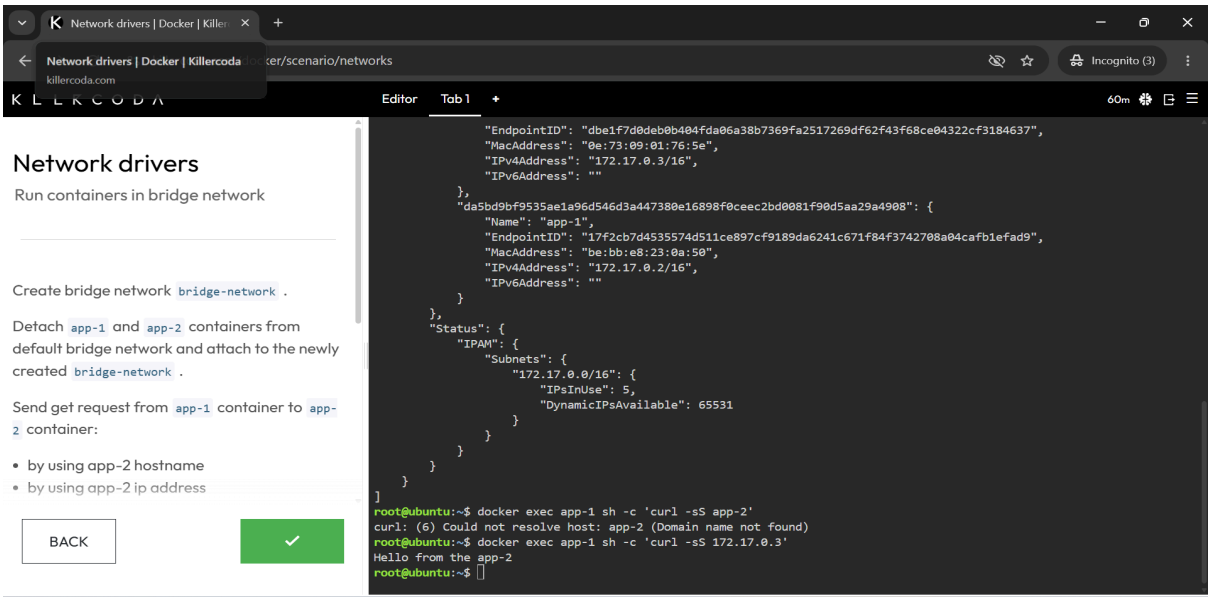


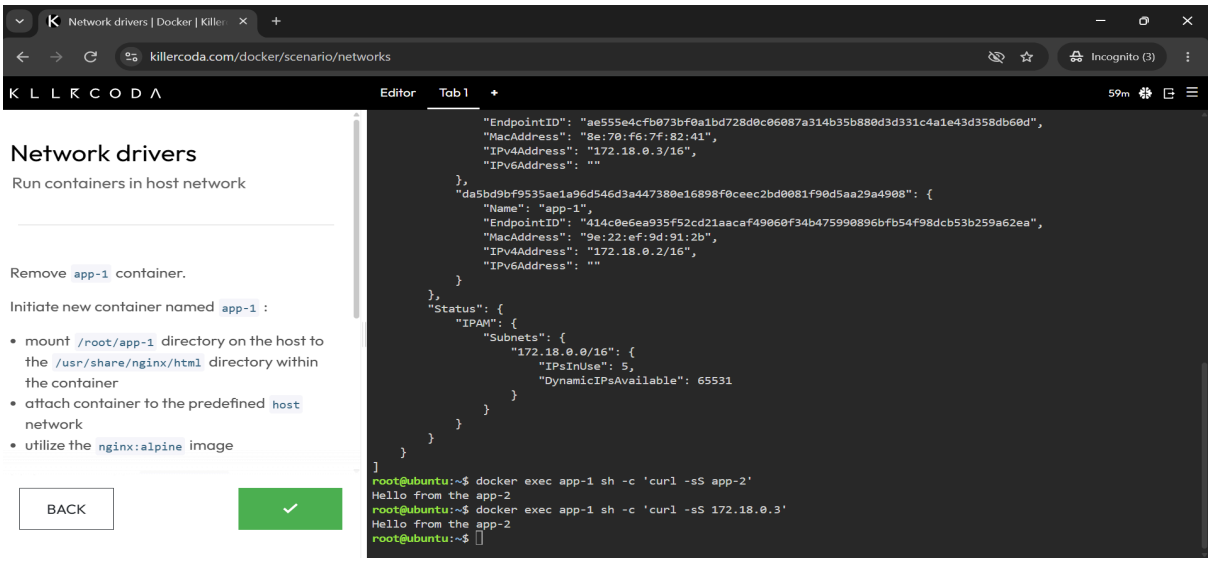


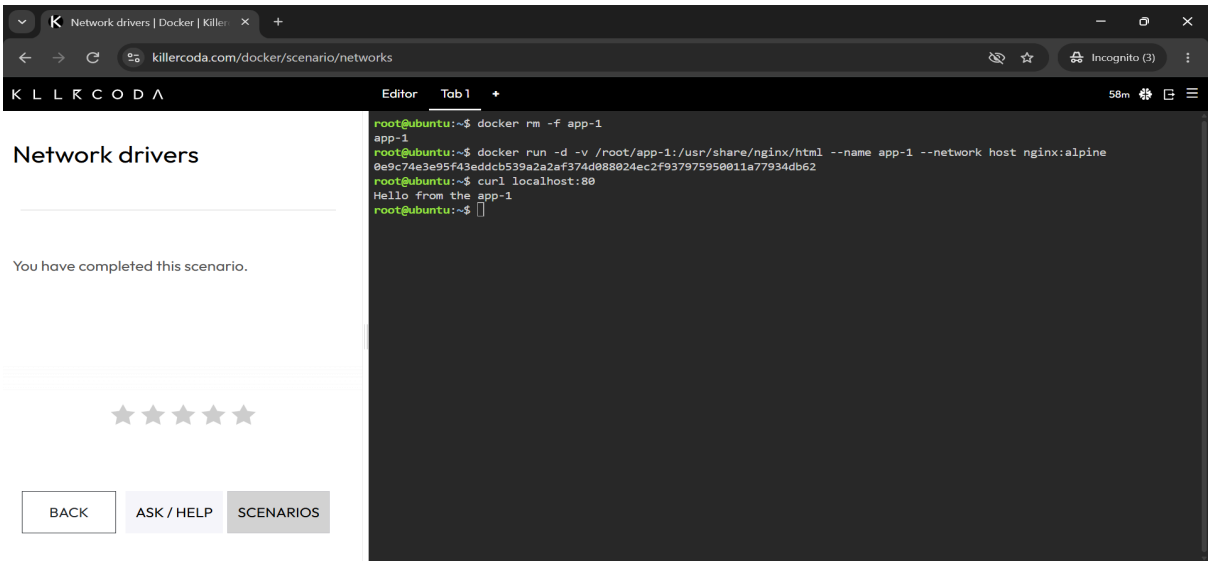

Using bind mounts | Docker | K
killercoda.com/docker/scenario/bind-mounts
Incognito (3)
KILLERCODA
Editor
Tab 1
59m
Using bind mounts
Change file in the bind mount
Add line "Added from the host" to the /root/app-data/index.html file on the host.
Send get request to localhost:80 .
Info
Tip
BACK
root@ubuntu:~$ docker exec sample-app sh -c "echo '<h1>Hello from the updated App</h1>' > /usr/share/nginx/html/index.html"
root@ubuntu:~$ curl localhost:80
<h1>Hello from the updated App</h1>
root@ubuntu:~$ docker rm -f sample-app
sample-app
root@ubuntu:~$ docker run -d -p 80:80 --mount type=bind,src=/root/app-data,target=/usr/share/nginx/html --name sample-app nginx:alpine
f24d8c9b4b109b7574f3f4712779fe39bbcf151aba7ffd1384676e30d9b3ec53
root@ubuntu:~$ curl localhost:80
<h1>Hello from the updated App</h1>
root@ubuntu:~$


Using bind mounts | Docker | K
killercoda.com/docker/scenario/bind-mounts
Incognito (3)
KILLERCODA
Editor
Tab 1
58m
Using bind mounts
You have completed this scenario.
BACK
ASK / HELP
SCENARIOS
root@ubuntu:~$ docker exec sample-app sh -c "echo '<h1>Hello from the updated App</h1>' > /usr/share/nginx/html/index.html"
root@ubuntu:~$ curl localhost:80
<h1>Hello from the updated App</h1>
root@ubuntu:~$ docker rm -f sample-app
sample-app
root@ubuntu:~$ docker run -d -p 80:80 --mount type=bind,src=/root/app-data,target=/usr/share/nginx/html --name sample-app nginx:alpine
f24d8c9b4b109b7574f3f4712779fe39bbcf151aba7ffd1384676e30d9b3ec53
root@ubuntu:~$ curl localhost:80
<h1>Hello from the updated App</h1>
root@ubuntu:~$ echo "Added from the host" >> /root/app-data/index.html
root@ubuntu:~$ curl localhost:80
<h1>Hello from the updated App</h1>
Added from the host
root@ubuntu:~$

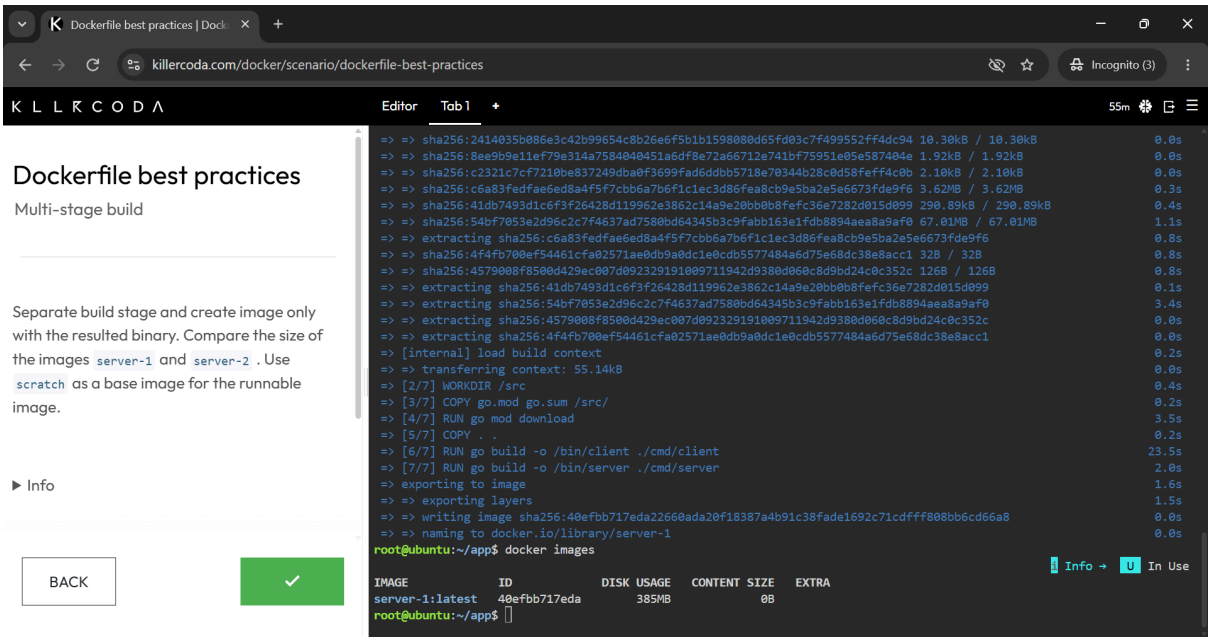
8. Network Drivers







9. Practice


Dockerfile best practices | Dock
killercoda.com/docker/scenario/dockerfile-best-practices
Incognito (3)
KILLERCODA
Editor
Tab 1
55m
Dockerfile best practices
Multi-stage build
Separate build stage and create image only with the resulted binary. Compare the size of the images server-1 and server-2 . Use scratch as a base image for the runnable image.
▶ Info
BACK
=> => sha256:2414035b086e3c42b99654c8b26e6f5b1b1598080d65fd03c7f499552ff4dc94 10.30kB / 10.30kB 0.0s
=> => sha256:8ee9b9e11ef79e314a7584040451a6df8e72a66712e741bf75951e05e587404e 1.92kB / 1.92kB 0.0s
=> => sha256:c2321c7cf7210be837249dba0f3699fad6ddbb5718e70344b28c0d58feff4c0b 2.10kB / 2.10kB 0.0s
=> => sha256:c6a83fedfae6ed8a4f5f7cbb6a7b6f1c1ec3d86fea8cb9e5ba2e5e6673fde9f6 3.62MB / 3.62MB 0.3s
=> => sha256:41db7493d1c6f3f26428d119962e3862c14a9e20bb0b8fefc36e7282d015d099 290.89kB / 290.89kB 0.4s
=> => sha256:54bf7053e2d96c2c7f4637ad7580bd64345b3c9fabb163e1fdb8894aea8a9af0 67.01MB / 67.01MB 1.1s
=> => extracting sha256:c6a83fedfae6ed8a4f5f7cbb6a7b6f1c1ec3d86fea8cb9e5ba2e5e6673fde9f6 0.8s
=> => sha256:4f4fb700ef54461cfa02571ae0db9a0dc1e0cdb5577484a6d75e68dc38e8acc1 32B / 32B 0.8s
=> => sha256:4579008f8500d429ec007d092329191009711942d9380d060c8d9bd24c0c352c 126B / 126B 0.8s
=> => extracting sha256:41db7493d1c6f3f26428d119962e3862c14a9e20bb0b8fefc36e7282d015d099 0.1s
=> => extracting sha256:54bf7053e2d96c2c7f4637ad7580bd64345b3c9fabb163e1fdb8894aea8a9af0 3.4s
=> => extracting sha256:4579008f8500d429ec007d092329191009711942d9380d060c8d9bd24c0c352c 0.0s
=> => extracting sha256:4f4fb700ef54461cfa02571ae0db9a0dc1e0cdb5577484a6d75e68dc38e8acc1 0.0s
=> [internal] load build context 0.2s
=> => transferring context: 55.14kB 0.0s
=> [2/7] WORKDIR /src 0.4s
=> [3/7] COPY go.mod go.sum /src/ 0.2s
=> [4/7] RUN go mod download 3.5s
=> [5/7] COPY . . 0.2s
=> [6/7] RUN go build -o /bin/client ./cmd/client 23.5s
=> [7/7] RUN go build -o /bin/server ./cmd/server 2.0s
=> exporting to image 1.6s
=> => exporting layers 1.5s
=> => writing image sha256:40efbb717eda22660ada20f18387a4b91c38fade1692c71cdfff808bb6cd66a8 0.0s
=> => naming to docker.io/library/server-1 0.0s
root@ubuntu:~/app$ docker images
I Info → U In Use
IMAGE ID DISK USAGE CONTENT SIZE EXTRA
server-1:latest 40efbb717eda 385MB 0B
root@ubuntu:~/app$